Fay Majid Elhassan, PMP

EPFL - Swiss Federal Technology Institute of Lausanne ,Lausanne,Switzerland
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Education

Doctoral Program in Computer and Communication Sciences (EDIC) — *Lausanne, Switzerland*
EPFL - Swiss Federal Technology Institute of Lausanne — *Sep 2025 – Present*
- Doctoral Assistant, Fellowship Program in Computer and Communication Sciences.
- **6% - 7%** Acceptance rate.

African Master of Machine Intelligence (AMMI), M.Sc in Mathematics, Major in Machine Learning. — *M'bour, Senegal*
African Institute of Mathematical Sciences (AIMS) — *Jan 2023 – Dec 2024*
- In partnership with **Google** and **Facebook (Meta)**.
- **6%** Acceptance rate.

Bachelor of Science Electrical and Electronic Engineering (Honours) — *Khartoum, Sudan*
University of Khartoum — *Oct 2014 – Oct 2020*

Research/Work Experience

Doctoral Assistant — *Lausanne, Switzerland*
Laboratory for Intelligent Global Health & Humanitarian Response Technologies – LiGHT Lab @ EPFL — *Sep. 2025 – Present*
- Massive Open Online Validation & Evaluation (MOOVE) — Research Lead
- Antibiogo Project — Research Lead in collaboration with Médecins Sans Frontières (MSF)

AI Technical Advisor — *Lausanne, Switzerland*
World Health Organization (WHO) — *Nov. 2025 – Present*
- Nominated Technical Advisor on the WHO-LEAD AI global innovation committee

Laboratory for Intelligent Global Health & Humanitarian Response Technologies – LiGHT Lab @ EPFL — *Lausanne, Switzerland*
Master's Research Intern — *Apr. 2025 – Aug. 2025*
- Conducted research on the Antibiogo Project in collaboration with Médecins Sans Frontières (MSF), under the supervision of Prof. Mary-Anne Hartley.

Max Planck Institute of Intelligent Systems — *Tubingen,Germany*
Research Engineer — *July 2024 - May 2025*
- Safety and efficiency-aligned learning group under Dr.Jonas Geiping and Dr.Antonio Orvieto for Project "Evaluating the promise of Language model watermarking for real-world cases".

Q-Parts — *Egypt/AlRiyadh-Remote*
Machine Learning Engineer — *June 2023 - May 2024*
- Developed and improved machine learning models and NLP techniques for a **PostgreSQL-based LLM chatbot**.
- Managed AI project pipelines, coordinating data analysis, preprocessing.

STEM-AWAY for Professional Training and Coaching — *Remote*
Principal Technical Lead - Machine Learning — *June 2023 - September 2023*
- Promoted to Principal Lead across diverse global teams, including Data Visualization, **AWS**, and **GPT**.

Dal Group — *Khartoum, Sudan*
Project Manager — *Nov 2022 - Dec 2023*
- Automated procurement, inventory, and invoicing for Sudan's **largest** agriculture project, **Project Green**, via a portal connecting suppliers and customers.

Business Intelligence Specialist — *April 2022 - Oct. 2022*
- Utilized forecasting models (**Arima**, **Sarimax**, **Prophet**) to predict sales for over **200** Dal food products, achieving **84%** accuracy with the Prophet model.

Data Scientist — *June 2021 - March 2022*
- Led digital transformation in the Marketing Department, applying **k-means clustering** for data analysis and creating inventory classification.

University of Khartoum
Teaching Assistant
Khartoum, Sudan
Feb 2021 - Dec 2022

- Taught Control System Theory and Computer-Aided Design to third and fourth-year university students, providing tutorials and individual assistance.

Publications

MoBayes: A Modular Bayesian Framework for Separating Reasoning from Language in Conversational Clinical Decision Support

Conference on Neural Information Processing Systems (NeurIPS) 2026 – Main Track *2026*

- Authors: Yusuf Kesmen*, Fay Elhassan*, Jiayi Ma*, Julien Stalhandske, Yena Chang, David Sasu, Alexandra Kulinkina, Akhil Arora, Lars Klein, Mary-Anne Hartley
- *Equal contribution (shared first authorship)
- Paper: `https://arxiv.org/abs/2604.20022`

I Can't Believe It's Not Safer: Preference–Safety Disassociation in Clinical LLM Evaluation

ICBINB Workshop, International Conference on Learning Representations (ICLR) 2026 – **Spotlight (Top 5%)** *2026*

- Authors: Fay Elhassan, David Sasu, Lars Henning Klein, Alexandra V. Kulinkina, Mary-Anne Hartley
- Spotlight presentation, top 5% of accepted papers
- Paper: `https://www.iclr.cc/virtual/2026/10014611`

Fully Open Meditron: An Auditable Pipeline for Clinical LLMs

GenAI4Health Workshop, Conference on Neural Information Processing Systems (NeurIPS) 2026 *2026*

- Authors: Xavier Theimer-Lienhard, Mushtaha El-Amin, Fay Elhassan, Sahaj Vaidya, Victor Cartier-Negadi, David Sasu, Lars Klein, Mary-Anne Hartley
- Paper: `https://arxiv.org/abs/2605.16215`

Can you Finetune your Binoculars? Embedding Text Watermarks into the Weights of Large Language Models

The 1st Workshop on GenAI Watermarking 2025 International Conference on Learning Representations *2025*

- Authors: Fay Elhassan, Niccolò Ajroldi, Antonio Orvieto, Jonas Geiping
- Paper: `https://arxiv.org/abs/2504.06446`

Comparison of deep reinforcement learning algorithms in enhancing energy trading in microgrids

2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE) *2021*

- Authors: Fay Elhassan, Moayad Elamin, Mahmoud A. Manzoul
- Paper: `https://ieeexplore.ieee.org/abstract/document/9429565`

Enhancing Energy Trading Between Different Islanded Microgrids: A Reinforcement Learning Algorithm Case Study in Northern Kordofan State

2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE) *2021*

- Authors: Fay Elhassan, Moayad Elamin, Mahmoud A. Manzoul
- Paper: `https://ieeexplore.ieee.org/document/9429584`

An Economic Evaluation of Islanded Microgrids Implementation in Northern Kordofan State

2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE) *2021*

- Authors: Fay Elhassan, Moayad Elamin, Mahmoud A. Manzoul
- Paper: `https://ieeexplore.ieee.org/document/9429680`

Projects

Multimodal Data Fusion and Contrastive Self-Supervised Learning for Enhanced Weather Forecasting with S4 Models

Remote

Research

Dec 2023 - Dec 2024

- Multimodal data fusion and contrastive self-supervised learning to align heterogeneous data sources. By incorporating the model **Structured State Space Sequence (S4)**
  Code:`https://github.com/FayElhassan/MultimodalRLSustainability`

Hybrid AI Recommendation Chatbot: AI Assistants

Remote

STEM-AWAY Company for Professional Training and Coaching

June 2023 - Oct 2023

- Developed a Hybrid AI Chatbot using **TF-IDF Vectorization** and **Cosine Similarity** for content and collaborative filtering recommendations.
  Code:`https://github.com/FayElhassan/HybridRecommendationChatbot`

DDPG & TD3 Implementation with PyBullet

Remote

AMMI Research

Sep 2023 - Oct 2023

- Developed **DDPG** and **TD3** algorithms using **JAX** for PyBullet's realistic robotic environments.
  Code:`https://github.com/FayElhassan/DDPG_IMP_PYBULLET`

PostgreSQL-Based LLM Chatbot

Egypt/Senegal

Q-parts

July 2023 - Oct 2023

- Developed a PostgreSQL-based chatbot using Langchain **LLM's** deployed in Streamlit, optimizing query generation, implementing caching, and enhancing database interaction efficiency.
  Code:`https://github.com/FayElhassan/Postgreschat`

Scientific Presentations

CaCTüS Scientific Symposium-Max Planck Institute for Intelligent Systems

Tuebingen,Germany

Embedding Subtle Watermarks in Large Language Models

24 Sep 2024

- Presenting the work of 3 month Cactus internship to Max Planck Institute network.

AMMI Scientific

Mbour,Senegal

AI for Sustainable Agriculture: Greenhouse Control Optimization

23 Aug 2023

- Presenting the application of various Reinforcement Learning (RL) algorithms for optimizing environmental conditions in greenhouses.

2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE)

Khartoum,Sudan

Enhancing Energy Trading Between Different Islanded Microgrids: A Reinforcement Learning Algorithm

28 Feb 2021

- Presenting the findings of the publication within the conference.

2020 International Conference on Computer, Control, Electrical, and Electronics Engineering (ICCCEEE)

Khartoum,Sudan

An Economic Evaluation of Islanded Microgrids Implementation in Northern Kordofan State

27 Feb 2021

- Presenting the findings of the publication within the conference.

Achievements/Certifications

2026	**Spotlight Paper (Top 5%),** ICLR 2026 ICBINB Workshop	*Rio de Janeiro, Brazil*
2024	**12th Africa and Middle East Gender Mainstreaming Awards,** Positive Role Model Finalist	*Middle East,Africa*
2024 ,2025	**CIFAR Inclusive AI Scholarship** CIFAR,	*Canada*
2021	**Above Traget,** Project Management Professional(**PMP**)	*Egypt*
2022	**Certified Application Associate,** SAP Analytics Cloud-Planning	*Sudan*
2022	**Certified Application Associate,** SAP Analytics Cloud	*Egypt*
2022	**Certified Application Associate,** Reporting, Modeling and Data Acquisition with SAP BW/4HANA 2.x	*Sudan*

Skills

Programming Languages
Python
SQL
PostgresSQL
R
C++
Nodejs
Tf.js

Libraries
Pytorch
Numpy
Matplotlib
Scikit-learn
Pandas
Jax

Tools/Techniques
Streamlit
Matlab
Git
Webscraping
PowerBI

Languages
English
Arabic
French

Leadership/Volunteerism

Institute of Electrical and Electronic Engineers Sudan *Khartoum, Sudan*
Chairman *2024 - present*

Sankofa Intiative:STEM What is in Hand? IEEE Smart Village
Cofounder/Chair of Communication and Documentation *2024- present*

IEEE Region 8 Student Activities Committee *Remote*
Publicity and Communication Coordinator *2024-2025*
- Past Chair of Digital Communications (2024- 2025)

Deep Learning Indaba (DLI) *Feb 2022 - August 2025*
- Organizer for the Main DLI and consultation on the INDABAX Sudan

Women in Machine Learning and Data Science-Khartoum Chapter *Khartoum, Sudan*
Cofounder *2021- present*